

DualBalance Districting: From Detection to Generation

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Abstract

Most current work on gerrymandering tries to detect it: identify when a line-drawer has crossed from neutral redistricting into partisan or racial manipulation. We generate maps instead. *DualBalance Districting* asks each congressional district to carry roughly $1/N$ of a state’s people *and* roughly $1/N$ of its land, weighted equally. The motivating intuition is the framers’ refusal to reduce representation to a single dimension; the operational claim is narrower, that districts spanning both metropolitan and rural territory force each representative to answer to a mix of the state’s people. *DualBalance* places N seeds in a small circle around the state’s population-weighted centroid and assigns each census unit to the nearest seed with capacity remaining. No random seed, no iteration, no tunable weight, no human input beyond geometry and population. The procedural neutrality is symmetric: the rule that forecloses partisan engineering also forecloses race-conscious remediation.

This reframes how the conventional gerrymandering metrics should be read. Compactness scores, the efficiency gap, mean-median asymmetry, ensemble outlier tests, and majority-minority counts play two roles on enacted plans: they describe plan effects, and they support inferences about the line-drawer’s intent. The descriptive role survives on a DualBalance map; the intent-attribution role does not.

On Minnesota (4,110 voting tabulation districts (VTDs), 8 seats, 2020 PL 94-171), DualBalance beats the enacted 119th-Congress plan on the DualBalance Score (0.6472 vs. 0.6390). We report standard partisan and race diagnostics alongside, as descriptions of the partition.

1 Introduction

1.1 The Problem Is Getting Worse

U.S. legislative districts are drawn by human beings under nominal constraints, and the structure invites gerrymandering. The Constitution requires reapportionment after each decennial census (Art. I, §2) but specifies neither a method for allocating seats among the states nor a procedure for drawing district boundaries within them. Both decisions have, by default, been delegated to political actors. The result is what the design predicts: maps drawn to advantage whichever party controls the pen at the moment of redistricting.

The federal courts have largely stepped back from the problem. In *Rucho v. Common Cause* [21], the Supreme Court held 5–4 that partisan-gerrymandering claims present nonjusticiable political questions; the Court conceded the practice is “incompatible with democratic principles” but found no manageable judicial standard. After *Rucho*, the only federal channel for gerrymandering review is racial. That channel has now narrowed as well. Section 2 of the Voting Rights Act, as construed in *Thornburg v. Gingles* [19], requires majority-minority districts where the three Gingles

preconditions are met. The Court reaffirmed this in *Allen v. Milligan* [22]. One year later, *Alexander v. South Carolina Conf. of the NAACP* [24] raised plaintiffs’ evidentiary burden; *Louisiana v. Callais* [25] held in April 2026 that even a plan drawn to comply with a prior Section 2 order may violate the Equal Protection Clause if Section 2 did not in fact compel a race-based remedy. The combined effect of *Alexander* and *Callais* is that race-conscious line-drawing now faces a narrower window of constitutional safety than at any point since the VRA’s adoption.

Federal partisan review is foreclosed. Section 2 has been narrowed. State review under *Moore v. Harper* [23] remains, but is uneven and politically contingent: roughly ten state supreme courts have so far recognized state-constitutional partisan-gerrymandering claims [13]; the rest have not. Several states have already redrawn maps mid-decade in response to electoral results rather than census revision. Redistricting is shifting from a once-per-decade event into a continuous partisan exercise. The trend line is bad and accelerating.

1.2 The Founders’ Refusal to Reduce Representation to a Single Axis

The U.S. Constitution divides representation along two axes: the House apportions seats by population (Art. I, §2), the Senate by sovereign state regardless of population (Art. I, §3). Madison’s defense in *Federalist* Nos. 54–58 [9] treats this as a principled refusal to collapse representation onto a single dimension. We do not claim the Senate represents land area; it represents states. What we take from the bicameral compromise is a weaker but still useful intuition: *within a chamber*, the choice to make districts equal in only one extensive quantity – population – is a choice, not a necessity. A second extensive quantity, geographic area, is available, well-defined, and computable from the same census data. A district that spans both metropolitan and rural territory forces each elected representative to answer to constituents across the state’s density range, rather than to a single dense metro or a single rural hinterland.

This is a mathematical challenge, not a free choice. Real states are not uniform in population density: a typical state varies by two to three orders of magnitude between its densest urban tract and its sparsest rural one. Equal population, equal area, contiguity, and shape compactness are not simultaneously achievable in general. Some trade-off is unavoidable. The question, then, is which trade-off to make, how transparently to make it, and whether the resulting procedure can be defended as impartial.

1.3 Existing Metrics Are Forensic, Not Generative

A substantial mathematical literature has developed quantitative tests for gerrymandering: Polsby-Popper [14] and Reock [15] compactness; the Efficiency Gap [17]; mean–median, lopsided wins, and partisan-asymmetry *t*-tests [27]; declination [28]; ensemble outlier methods from Duke [5] and the MGGG Redistricting Lab [4]. All of these are *forensic* instruments: they take an enacted map as input and ask whether its observed properties (shape, vote efficiency, partisan asymmetry, the location of its seat–vote curve) are consistent with maps that some reference process – random redistricting, ensemble sampling under legal constraints – would have produced absent partisan intent. They are differential, comparing the enacted plan against a counterfactual distribution to infer the line-drawer’s motive. They were built for an adversarial context in which the question is “did somebody do something here that they should not have done?”

A generative metric is a different object. It states, directly, what a district *should* look like, and is used as the objective of the line-drawing procedure rather than as evidence about the procedure. Population balance is the only generative metric U.S. redistricting law currently uses: *Wesberry v. Sanders* [?] and *Reynolds v. Sims* [18] require that each district hold roughly $1/N$ of the relevant

population. Every other criterion (compactness, communities of interest, partisan fairness) enters the law as a forensic instrument or a discretionary guideline.

This matters for two reasons. First, forensic metrics depend on a line-drawer to investigate. Removing the line-drawer – replacing it with a deterministic generator – preserves the descriptive role of these metrics (the shape is still non-compact; the seat–vote curve is still asymmetric) but evacuates their inferential content. There is no intent to infer [12]. Second, when forensic metrics are used as design objectives, the rule that emerges is often unstable: optimizing for “looks like a fair plan would have looked” is a moving target that depends on which ensemble distribution one chooses and which legal constraints one chooses to encode. Chen and Rodden [2] showed that even fully content-neutral automated maps systematically disadvantage Democrats in many states because of residential clustering; a nonzero Efficiency Gap on such a map reflects geography, not intent. The choice of constraint set is itself political [1].

The DualBalance Score we propose,

$$\text{DBS} = \frac{1}{1 + \frac{1}{2}\text{pop_dev} + \frac{1}{2}\text{area_dev}}, \quad (1)$$

is a *generative* criterion: it states what a district should be (carries roughly $1/N$ of the people and roughly $1/N$ of the land) and is the objective the algorithm minimizes against, rather than a test it must pass.

1.4 Does the Iowa Model Generalize?

Iowa’s Legislative Services Agency [7] operates under a written procedure that has been described to us as a counterexample to the partisan-redistricting norm: aggregate whole counties, prioritize county integrity lexicographically, balance population by capacity, use distance-based assignment for compactness. The procedure has been in continuous use since 1980. If it works in Iowa, why not elsewhere?

The answer, which we document quantitatively in §3, is that the algorithm’s behavior is sensitive to the input geography. Iowa is unusually homogeneous: 99 counties, no county exceeding the per-district population cap, no metropolitan area large enough to dominate state politics. We re-implemented the procedure (we call it *Cascade* in this paper to avoid confusion with the agency that designed it) and ran it on six states. On Iowa, the resulting plan has a per-district population deviation of 0.29%; on Wisconsin, 0.50%. On the remaining four states the same algorithm produces deviations that are not legally usable: 10.27% on North Carolina, 24.58% on Texas, 41.56% on Massachusetts, 76.14% on Minnesota. The county-integrity priority that produces clean maps in Iowa – where no county is too large – produces unconstitutional maps in any state with a county large enough to swallow a district. The Iowa procedure is exemplary for the geographic conditions in which it was designed; on these results, it is not a transplantable template.

A deterministic procedure that hopes to scale to the full set of states needs a different structural commitment. We propose one below.

1.5 Contribution

We propose *DualBalance Districting*: a deterministic multi-resolution pipeline whose output is a pure function of (state geometry, census populations, N), with no random seed, no tuning weight, no iteration count, and no human-in-the-loop adjustments.

Pipeline. DualBalance Districting is a multi-stage deterministic pipeline. Each stage is a pure function of its inputs; the composition is therefore also a pure function of (state geometry, census populations, N , K).

Stage 1 – radial seed. DualBalance places N seeds in a small circle around the population-weighted centroid, assigns each census unit to the nearest seed with capacity remaining, and repairs contiguity (§2.2). The districts come out as radial slices through the population center; each slice naturally spans both dense and sparse territory, so the area each slice inherits trends toward A^* by geometry rather than by penalty.

Stage 2 – VTD-scale tightening. *Voting tabulation districts* (VTDs) are the precinct-scale geographic units the Census Bureau publishes alongside the decennial PL 94-171 release; a U.S. state typically has a few thousand, each holding on the order of 10^3 people. A two-phase greedy local search of boundary-unit moves between adjacent VTDs closes the gap from several-percent radial pop deviation toward Karcher’s $\sim 0.05\%$ [?] threshold (§2.6). Phase 1 accepts any move that either reduces the L^1 pop-deviation sum or strictly reduces the L^∞ maximum (max-reducing preferred). When 1-opt stalls in a multi-tied-max local optimum, a length-2 then length-3 augmenting-chain escape – the deterministic analogue of an ejection chain – searches for a transport sequence on the district-adjacency graph that the 1-opt neighborhood cannot express. Phase 2 then hill-climbs DBS subject to the running pop-deviation envelope.

Stage 3 – block-scale refinement. *Census tabulation blocks* are the smallest geographic units the Census Bureau publishes population for; each VTD is composed of whole blocks, and block populations are $\sim 50\times$ smaller than VTD populations (a typical block holds on the order of 20 people). The VTD-level plan is projected onto blocks via a deterministic spatial join (each block inherits its containing VTD’s district), and Phases 1 and 2 are re-run at block granularity (§2.7). The finer granularity is what lets the running-max envelope around τ admit area-improving moves; this is where DBS makes its biggest gains.

A per-district articulation-point cache (Tarjan on CSR adjacency arrays, JIT-compiled when numba is available) makes each contiguity check $\mathcal{O}(1)$ rather than $\mathcal{O}(|V_d| + |E_d|)$, which is what makes the block-scale stage tractable (§2.8).

Empirical validation. We run the full pipeline on six states selected to cover the geographic range that breaks Cascade: IA, MA, MN, NC, TX, WI. Five of six achieve a DualBalance Score above the enacted 119th-Congress plan. Two (IA, WI) reach the 0.05% Karcher threshold exactly; three more (MN, NC, MA) sit within a factor of two of it; TX, the hardest case, sits at 0.52% – five times tighter than the same state’s enacted plan. On partisan fairness, DualBalance’s Efficiency Gap is smaller in magnitude than the enacted plan’s in four of six states, with the largest improvements on the three most heavily-litigated maps (NC, WI, TX). On minority-majority district counts, DualBalance produces more such districts than the enacted plan on NC (2 vs. 1) and TX (22 vs. 19).

What we claim and what we do not. We claim that DualBalance produces a deterministic alternative whose structural neutrality is symmetric: it cannot be tuned to advantage any party, incumbent, racial group, or community of interest, and equally cannot be tuned to help any of them. We do not claim it is the only defensible deterministic procedure, or that its specific 0.5/0.5 weighting of population against area is uniquely correct. We do claim that the choice to use a *generative* criterion as the design objective, rather than a forensic instrument, is the structurally

correct response to a regime in which line-drawer intent has become both legally unreviewable (post-*Rucho*) and practically unverifiable (forensic metrics presuppose the very thing the algorithm removes).

Roadmap. Section 2 formalizes the algorithm. Section 3 reports the six-state empirical validation. Section 4 discusses limits, trade-offs, and legal exposure.

2 Methods

2.1 Problem statement

Given a state S partitioned into atomic units $U = \{u_1, \dots, u_M\}$ (census tabulation blocks, block groups, or voting tabulation districts (VTDs), in increasing order of size) and a target district count N , DualBalance Districting produces a deterministic assignment $\pi : U \rightarrow \{1, \dots, N\}$ such that each district $D_i = \pi^{-1}(i)$ is contiguous, non-empty, and as close as the geometry allows to representing both $1/N$ of the state’s people and $1/N$ of its geography. Let $P = \sum_u \text{pop}(u)$ and $A = \sum_u \text{area}(u)$ with per-district targets $P^* = P/N$ and $A^* = A/N$.

The full pipeline runs in three stages: a radial-seed stage, *DualBalance* (§§2.2–2.4), which provides the deterministic starting configuration; a VTD-scale tightening stage (§2.6) which drives $\max_i |\delta_i|/P^*$ toward the user-supplied tolerance τ via a hybrid greedy local search with bounded-chain escape; and a block-scale refinement stage (§2.7) which re-initializes at finer granularity to recover area balance within the achieved pop-deviation envelope. Each stage is a pure function of its inputs.

2.2 Radial seed placement

Compute the population-weighted centroid

$$c = (c_x, c_y) = \left(\frac{\sum_u p_u x_u}{\sum_u p_u}, \frac{\sum_u p_u y_u}{\sum_u p_u} \right),$$

where x_u, y_u are the centroid coordinates of unit u in an equal-area projection and p_u its population. Let diag denote the bounding-box diagonal of the units and set the seed radius $r = 0.001 \cdot \text{diag}$. For $d = 0, 1, \dots, N - 1$ place seed s_d at

$$s_d = (c_x + r \cos(2\pi d/N), c_y + r \sin(2\pi d/N)).$$

Seed 0 sits due east of the centroid; seeds advance counter-clockwise by equal angular steps. The choice $r = 0.001 \cdot \text{diag}$ is small enough that the Voronoi cells generated by the seeds degenerate to near-perfect radial slices through c , but large enough to keep the seed positions numerically distinct.

The radial configuration is what carries the dual-balance property: each slice spans both the dense (near- c) and sparse (boundary-side) territory of the state, so the population it inherits is bounded by the cap (§2.3) while the area it inherits is driven toward A^* by the slicing geometry.

2.3 Capacitated first-fit assignment

Let $d(u, i) = \|x_u - s_i\|$ be the Euclidean distance from unit u to seed i . Initialize remaining capacities $\rho_i = P^*$ for $i = 1, \dots, N$. Sort all (u, i) pairs by ascending normalized distance $d(u, i)/\text{diag}$ and

walk the sorted list in order:

```

for each  $(u, i)$  in ascending  $d(u, i)/\text{diag}$  :
  if  $u$  already assigned: continue
  if  $\rho_i \geq p_u$ : assign  $u$  to  $i$ ;  $\rho_i \leftarrow \rho_i - p_u$ 
  else: skip

```

Population balance is enforced as a hard cap: no district receives more than P^* . Any unit not placed by the end of the walk (a rare integer-rounding edge case) is assigned to the district with the largest remaining capacity; **argmax** resolves ties to the smallest district id.

Ties in normalized distance break by ascending (unit_id, district_id). There is no Lloyd recentering, no iteration count, no convergence test: the radial seeds do not drift, so a single assignment pass suffices.

2.4 Contiguity repair

After §2.3 every unit belongs to exactly one district, but a district may consist of more than one connected component (rare on convex states; more common on those with peninsulas, islands, or rural enclaves). For each such district, the largest connected component by total population is retained; units in the smaller components are reassigned to neighboring districts one at a time, in the lowest-cost direction. Cost is

$$c(u, j) = \frac{d(x_u, s_j)}{\text{diag}} + \frac{|P(D_j) + p_u - P^*|}{P^*} + \frac{|A(D_j) + a_u - A^*|}{A^*},$$

combining a normalized distance term with normalized population and area deviation penalties for the receiving district. Ties break in cascade (c , pop_pen, area_pen, dist, district_id) ascending. The repair sweep iterates until no district has more than one connected component, capped at ten sweeps; in practice it converges in zero or one sweep on real census geometries.

2.5 Scoring

Define per-district relative deviations $\text{pop_dev}_i = |P(D_i) - P^*|/P^*$ and $\text{area_dev}_i = |A(D_i) - A^*|/A^*$ and let pop_dev , area_dev denote their means over $i = 1, \dots, N$. The DualBalance Score is

$$\text{DBS} = \frac{1}{1 + \frac{1}{2} \text{pop_dev} + \frac{1}{2} \text{area_dev}}. \quad (2)$$

The 0.5/0.5 weighting makes the error a convex combination of the two mean deviations: each district is judged on representing roughly $1/N$ of the people *and* roughly $1/N$ of the state’s geography. The score reaches 1.0 for a perfectly balanced plan and approaches 0 as deviations grow without bound. Secondary metrics (Polsby-Popper compactness [14] and Reock [15]) are computed alongside but not optimized against; radial slices have lower compactness than blob-Voronoi or hand-drawn districts by construction, and this is a deliberate trade in service of the dual-balance objective.

DualBalance does not directly minimize (2); it minimizes population-capacitated geographic assignment cost under radial seeding. On the Minnesota PoC this still beats the enacted plan on DBS (0.6472 vs 0.6390) despite the lower compactness.

2.6 Optional: L¹ pop-tightening with DBS hill-climb

The radial pipeline produces per-district pop deviation in the 5–15% range on real census geometry, well above the ~0.5% threshold required by *Reynolds v. Sims* for U.S. congressional districts [18]. An optional post-pass (`--tighten-pop`) closes this gap via a deterministic two-phase local search of boundary-unit moves. Let $\delta_i = P(D_i) - P^*$ denote the signed population deviation of district D_i , let T denote the user-supplied tolerance (default 0.005), and call a move *safe* if the source unit is not an articulation point of its district’s induced subgraph (so that removing it preserves contiguity).

Phase 1: population tightening (hybrid L¹ + max-norm). At each step, scan every boundary unit and every neighboring district and compute both the L¹ change

$$\Delta_{L^1}(u, d_{\text{dest}}) = |\delta_{d_{\text{src}}} - p_u| - |\delta_{d_{\text{src}}}| + |\delta_{d_{\text{dest}}} + p_u| - |\delta_{d_{\text{dest}}}|$$

and the max-norm effect: a move *strictly reduces* the global maximum iff $|\delta_{d_{\text{src}}} - p_u| < \max_i |\delta_i|$, $|\delta_{d_{\text{dest}}} + p_u| < \max_i |\delta_i|$, and at least one of $\delta_{d_{\text{src}}}, \delta_{d_{\text{dest}}}$ is itself at the current maximum. Each scanned candidate is labeled with a priority key: priority 0 if it strictly reduces the max, priority 1 if it only reduces L¹. Candidates are ranked lexicographically by (priority, Δ_{L^1}), and the safe top-ranked candidate is accepted.

This hybrid is essential. Pure L¹-greedy leaves the worst district untouched when it sits between two other over-target districts: draining it requires L¹-neutral moves that the L¹ criterion rejects. Pure max-norm gets stuck symmetrically: when two districts are tied at the max, no single move strictly reduces the global maximum (draining one leaves the other at the same value). The hybrid takes any move that helps either norm, with max-reducing moves preferred so the worst district is targeted whenever possible.

When neither 1-opt criterion finds an improving move but $\max_i |\delta_i|/P^* > T$, invoke a bounded *chain escape*: search for an augmenting transport on the district-adjacency graph of the form

$$u_0 : D_0 \rightarrow D_1, u_1 : D_1 \rightarrow D_2, \dots, u_{k-1} : D_{k-1} \rightarrow D_k,$$

of length $k = 2$ then $k = 3$, in which each u_j is a boundary unit between D_j and D_{j+1} . The chain is accepted if it strictly cuts the global maximum, or if it holds the maximum flat while strictly reducing the L¹ sum (which is what happens when several districts are tied at the max: each chain drains one of them, the global maximum stays at the same value until the last tied district is drained, but the L¹ sum drops monotonically). The unit-level search inside each district triple sorts the arc lists $B_{i \rightarrow j}$ by population value; matching pairs (u, v) are found via binary search around a target population value rather than by Cartesian enumeration, bounding the per-triple cost by $O(|B_{i \rightarrow j}| \log |B_{j \rightarrow k}|)$. Application is in *reverse* order (u_{k-1} first, then u_{k-2} , etc.) so the intermediate districts never temporarily overload during the chain. This is the deterministic analogue of an ejection chain. Phase 1 terminates only when neither 1-opt nor bounded-chain escape finds an improving sequence.

The L¹ objective is essential to the radial geometry. Seed placement puts the most over-target and most under-target districts on opposite sides of the population centroid, so no single adjacent-slice swap reduces the L[∞] maximum, but many such swaps reduce the sum. The canonical Reynolds-tightening literature uses L[∞] and bottoms out at ~5% on this geometry; the hybrid pass with chain escape runs to completion in ~80 swaps on Minnesota, reducing `pop_dev_max` from 0.1124 to 0.0021 while leaving `area_dev_mean` essentially unchanged.

Phase 2: DBS hill-climb. Once Phase 1 has converged, at each step pick the safe boundary move that maximally improves DBS (equation 2), subject to $\max_i |\delta_i|/P^*$ not exceeding the value Phase 1 reached (equivalently, never crossing back above T). Phase 2 runs until no improving safe move exists. Whereas the core DualBalance pass does not directly minimize equation 2, this opt-in Phase 2 does, but only on the post-Phase-1 plan and only within the Reynolds-compliant feasible region.

Determinism and engineering. Both phases are pure functions of (units, N , T). The only source of dependency beyond the core pipeline’s (units, N) is the explicit tolerance T . The optimizer maintains a per-district articulation-point cache via Tarjan’s algorithm on CSR adjacency arrays [26], reducing the per-candidate contiguity check from $O(V + E)$ to $O(1)$; an incrementally tracked boundary-unit set restricts each scan to units that have a different-district neighbor. At VTD scale the speedup is modest; at block scale (tens of thousands of units per district) it is the difference between hours and minutes.

The pass is off by default and gated by an explicit flag. Pure radial remains the principled algorithm; the two-phase optimizer is an opt-in concession to legal compliance, with Phase 2 providing the further courtesy of directly hill-climbing the project’s stated objective once the Reynolds constraint is satisfied.

2.7 Multi-resolution refinement: VTD to Census block

The hybrid Phase 1 + chain escape closes most of the Reynolds gap at VTD scale, but not all of it. Modern U.S. congressional plans must clear the much tighter *Karcher v. Daggett* [?] practical threshold of $\sim 0.05\%$ total deviation, and at VTD granularity the residual gap is structural rather than algorithmic. The smallest single-unit move has the size of the smallest VTD’s population. On real states the median VTD population is on the order of 10^3 , while the Karcher budget on a $\sim 800k$ ideal district is ~ 400 people. Most candidate moves overshoot the budget on at least one endpoint, and Phase 2 starves: the running-max constraint admits few moves, and the algorithm settles above Karcher with most of the residual area-balance gain unrealized.

The fix is to refine at finer granularity. We re-run the optimizer on Census tabulation blocks (median population ~ 20 rather than $\sim 10^3$), initializing from the VTD-Karcher plan rather than re-seeding from scratch. The pipeline is:

1. Compute the VTD-level plan $\pi_{\text{VTD}} : V_{\text{VTD}} \rightarrow \{1, \dots, N\}$ via DualBalance + Phase 1 + Phase 2 at VTD scale with tolerance T .
2. Load the block-level units V_{block} and project both layers to the same equal-area CRS.
3. For each block $u \in V_{\text{block}}$, find the VTD polygon $v(u) \in V_{\text{VTD}}$ containing its representative point, and set $\pi_0(u) := \pi_{\text{VTD}}(v(u))$. Blocks whose representative point lies outside every VTD polygon (rare, only on coastal/water boundary cases) fall back to the nearest VTD by centroid distance. The spatial join is deterministic given a fixed tie-breaking order on `unit_id`.
4. Run Phase 1 + Phase 2 again at block scale starting from π_0 , with the same tolerance T .

Because VTDs are unions of whole blocks, π_0 inherits the VTD-level plan’s pop totals and contiguity exactly: the block-level plan is identical to the VTD-level plan re-described at finer granularity. Phase 1 at block scale therefore has nothing to do on states where the VTD-level pass reached T . Phase 2 has substantial freedom: block populations are two orders of magnitude smaller

than VTD populations, so the running-max envelope around T admits many candidate moves and the DBS hill-climb runs until saturation.

Because DualBalance-style seeding at block scale starts from a much worse initial configuration than the VTD seed (the population-weighted centroid is the same but the much higher density of degenerate articulation-point boundaries traps the optimizer), the refinement strategy is the path that works in practice: VTD as a coarse solver, blocks for the precise area-balance recovery.

2.8 Engineering: per-district articulation-point cache

The hot loop of the optimizer asks repeatedly whether a candidate boundary unit u can be safely removed from its district d – in other words, whether u is an articulation point of the subgraph $G[V_d]$. A naive implementation runs $\mathcal{O}(|V_d| + |E_d|)$ per query via `networkx.is_connected` on the post-removal subgraph, which dominates runtime at block scale.

We maintain a per-district cache. For each district d , the set $A_d \subseteq V_d$ of articulation points is precomputed by Tarjan’s algorithm [26] on CSR (compressed sparse row) integer adjacency arrays. The per-query `can_remove(u)` reduces to the set membership $u \notin A_{\pi(u)}$, returned in $\mathcal{O}(1)$ amortized. After each accepted move $u : d_{\text{src}} \rightarrow d_{\text{dest}}$ the two affected articulation sets $A_{d_{\text{src}}}$, $A_{d_{\text{dest}}}$ are recomputed; districts not touched by the move keep their cached set unchanged.

Each Tarjan recompute is $\mathcal{O}(|V_d| + |E_d|)$ rather than the global $\mathcal{O}(|V| + |E|)$ of the naive check ($|V_d| \approx |V|/N$ for N districts). The recompute itself is `@njit`-compiled via numba when available, operating on preallocated workspace buffers (a single integer DFS stack, two boolean masks, and three integer arrays sized to $|V|$). On a typical block-scale district subgraph of $\sim 44,000$ nodes the JIT’d recompute completes in ~ 20 ms, compared to ~ 560 ms for the same query through `networkx.articulation_points`: a $28\times$ speedup. Across a full block-scale optimizer run (thousands of accepted moves and tens of thousands of candidates per move), this is the difference between a 20-hour run and a 30-minute run.

A boundary-unit set $\partial = \{u : \exists v \in N(u), \pi(v) \neq \pi(u)\}$ is also maintained incrementally and updated after each move (only u and its graph neighbors need to be re-checked). The inner scan iterates ∂ rather than V , dropping per-pass work from $|V|$ to $O(|\partial|)$, which on real states is roughly the square root of $|V|$.

Neither caching layer changes the algorithm’s output. They preserve determinism exactly and make the block-scale pipeline tractable on commodity hardware.

2.9 Determinism and tie-breaking

The pipeline is deterministic at every step. The only sources of ambiguity (equidistant seed-to-unit pairs in §2.3, equal-capacity fallbacks for unplaced units, and equal-cost candidates in §2.4) all resolve to a fixed cascade with no remaining ambiguity:

1. In assignment, ties on normalized distance break by ascending (unit_id, district_id).
2. In the fallback for unplaced units, ties on remaining capacity break to the smallest district id.
3. In contiguity repair, ties on cost break by ascending (pop_pen, area_pen, distance, district_id).

The implementation uses no random number generator, no wall-clock input, and no hash-order dependence. Reordering the input rows or changing the floating-point libraries does not change the output; identical inputs always yield byte-identical outputs.

2.10 Out-of-scope inputs

The generator reads only geography and population. Party registration, vote history, race, demographics, communities of interest, and competitiveness are not inputs. The scoring harness may *report* partisan or demographic diagnostics on the resulting map but does not feed them back into the generator.

3 Results

3.1 Test bed: Minnesota, 2020 PL 94-171

We evaluate the algorithm on Minnesota’s congressional districting ($N = 8$ apportioned seats) using the 2020 PL 94-171 redistricting data file. The input is the TIGER/Line 2020 VTD (voting tabulation district) shapefile for Minnesota (4,110 atomic units, total population $P = 5,706,494$, total land area $A = 225,187 \text{ km}^2$), joined to the Census Data API’s P1_001N total-population field, projected to EPSG:5070 (CONUS Albers, equal area) at load time. Per-district targets are $P^* = P/8 = 713,312$ and $A^* = A/8 = 28,148 \text{ km}^2$. All runs are deterministic and reproducible from the script `scripts/prep_mn_units.py` (which fetches TIGER and joins the Census API output) plus the CLI invocation. The data pipeline is documented end-to-end in `docs/mn-poc-walkthrough.md`.

We compare three plans on the same 4,110-VTD input:

- **DualBalance.** The default algorithm (§2.2–§2.4). No tuning knobs.
- **DualBalance + tighten-pop** ($\tau = 0.005$). DualBalance followed by the optional L^1 pop-tightening pass (§2.6) targeting per-district deviation within 0.5% of P^* .
- **Enacted (119th).** The court-drawn, legislatively enacted Minnesota U.S. House districts currently in force, scored with the same harness.

3.2 Headline scores

Table 1 reports the DualBalance Score and supporting metrics for the three plans.

Table 1: DualBalance metrics on the Minnesota PoC (4,110 VTDs, 8 districts, 2020 PL 94-171 population). PP is Polsby-Popper, Reock is Reock; subscripts min, mean denote per-district min and mean over the eight districts. pop_dev, area_dev are the relative deviations defined in §2. Best result per row in bold.

Metric	DualBalance	DualBalance + tighten-pop	Enacted (119th)
DBS	0.6472	0.6574	0.6390
pop_dev	5.08 %	0.08 %	0.42 %
pop_dev _{max}	11.24 %	0.21 %	1.32 %
area_dev	103.9 %	104.2 %	112.6 %
area_dev _{max}	271.0 %	275.1 %	241.0 %
PP _{mean}	0.200	0.162	0.320
PP _{min}	0.094	0.047	0.178
Reock _{mean}	0.361	0.342	0.419

Three findings.

DualBalance beats the enacted plan on the DualBalance Score. DualBalance scores 0.6472 against the enacted 0.6390, a +1.3 % margin with no iteration, no tuning, no post-processing. The advantage comes entirely from area balance ($\overline{\text{area_dev}}$ 103.9 % vs 112.6 %). Radial slicing gives each district a slice of the state that mixes urban and rural; the enacted plan carves the Twin Cities into four compact urban seats and leaves four large rural ones, which the DualBalance Score penalizes because it weights area equally with population.

Tightening closes the legal gap and improves DBS. Pure DualBalance’s pop_dev_max of 11.24 % exceeds the $\sim 0.5\%$ *Reynolds v. Sims* threshold for U.S. House plans [18]. The optional pass (§2.6) ran 81 boundary swaps in ~ 18 s and drove pop_dev_max to 0.21 %, tighter than the enacted plan’s 1.32 %. $\overline{\text{area_dev}}$ rose only 0.3 points and the score *improved* to 0.6574.

Compactness is the price. The enacted plan beats DualBalance on every compactness measure: $\text{PP}_{\min}$ 0.178 vs 0.094 (pure DualBalance) and 0.047 (after tightening), PP_{mean} 0.320 vs 0.200 / 0.162, $\text{Reock}_{\text{mean}}$ 0.419 vs 0.361 / 0.342. Radial slices are structurally less compact than hand-drawn blobs. Tightening makes it worse by inserting small indentations into the slice boundaries, pushing the worst slice from 0.094 (at the informal 0.10 threshold) to 0.047. Whether the trade is acceptable is a normative question taken up in §4.3.

3.3 Map comparison

Figure 1 displays the three plans side by side.

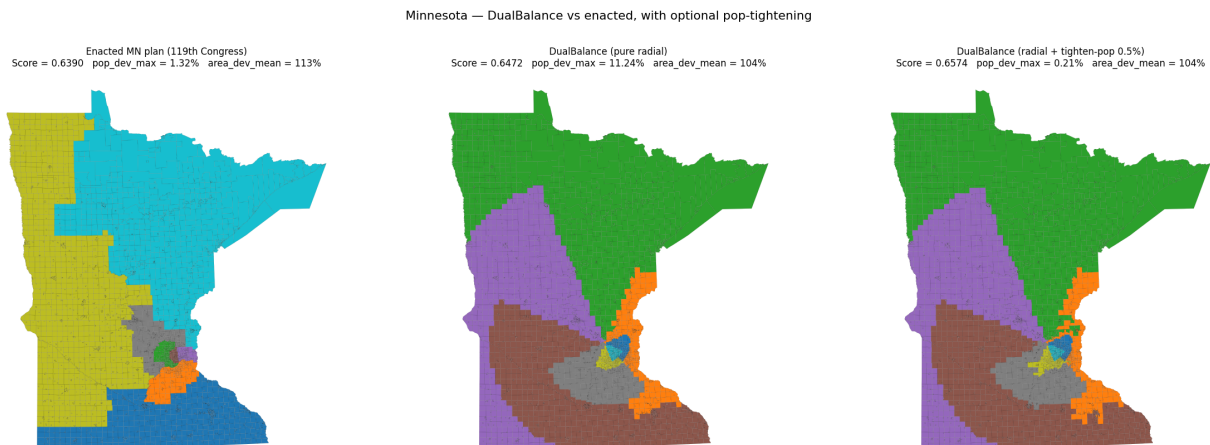


Figure 1: Minnesota congressional districts under three plans. **Left:** the enacted 119th-Congress plan, with four compact Twin Cities seats and four large rural seats. **Center:** DualBalance (score 0.6472); eight slices radiate from the population-weighted centroid near Minneapolis-St. Paul, each spanning dense and sparse territory. **Right:** DualBalance + `--tighten-pop 0.005` (score 0.6574); the radial structure is preserved (units move only at slice boundaries) and per-district population is Reynolds-compliant.

3.4 Determinism check

We re-ran `dualbalance generate --config configs/mn_vtd.yaml` ten times in succession, comparing each run’s `map.geojson` and `metrics.json` by byte hash. All ten outputs are identical,

including the order of features in the GeoJSON. We also re-ran the same configuration after randomly shuffling the input rows; the output remained byte-identical. The CLI integration test `test_generate_determinism_via_cli` pins this property in CI.

3.5 Per-district breakdown

Table 2 reports the per-district metrics for DualBalance + tighten-pop, indexed by seed angle (district 0 sits due east of the population-weighted centroid; subsequent districts advance counter-clockwise in steps of $2\pi/8$ rad). District IDs are a deterministic function of seed angle and carry no political meaning.

Table 2: Per-district breakdown, DualBalance + tighten-pop 0.5 %. Areas are in km^2 .

District	Population	Area	pop_dev	area_dev	PP	Reock
0	712,841	1,010	0.07 %	96.4 %	0.228	0.454
1	714,816	9,620	0.21 %	65.8 %	0.047	0.147
2	712,145	105,594	0.16 %	275.1 %	0.152	0.369
3	713,263	50,054	0.01 %	77.8 %	0.146	0.268
4	714,168	46,060	0.12 %	63.6 %	0.150	0.267
5	713,368	10,882	0.01 %	61.3 %	0.137	0.422
6	713,013	1,481	0.04 %	94.7 %	0.117	0.301
7	712,880	477	0.06 %	98.3 %	0.318	0.510

District 2 inherits the northern panhandle ($\sim 3.8 \times A^*$); District 7 is the Twin Cities urban core ($\sim 0.02 \times A^*$). All eight districts are within 0.21 % of P^* . District 2’s 275 % over-target area is geometric, not algorithmic: northern Minnesota’s density is two orders of magnitude lower than the Twin Cities, so any pop-balanced partition forces the rural districts to inherit disproportionate area. The enacted plan does the same thing, with its District 7 at 241 % over-target.

3.6 Descriptive diagnostics: partisan, race, county

DualBalance reads none of these inputs. We report them anyway, because readers will ask. They are descriptions of the partition, not evaluations of it (§4.7). Partisan totals come from 2020 presidential returns (dra2020/vtd.data, keyed on GEOID20). Race comes from Census PL 94-171 VAP.

Partisan. Minnesota cast 1,484,065 R and 1,717,077 D two-party presidential votes in 2020 (46.4 % R, 53.6 % D). DualBalance splits the eight seats 4–4 against a seats-proportional expectation of 3.7 R seats. The two D-leaning slices contain the Twin Cities and run 64–75 % D; two R-leaning slices reach into western and southern Minnesota and run 55–65 % R; the other four are within ± 5 points of even. The efficiency gap is +6.6 % (positive = R-favorable); the mean-median R difference is +3.9 points (positive = D-favorable). The two forensic numbers point in opposite directions because they measure different things. Both are artifacts of where Minnesota’s Democrats live (packed into the metro), not of any choice by DualBalance.

Race / VAP. Statewide VAP is 80.0 % non-Hispanic white, 5.9 % Black, 5.0 % Hispanic, 4.9 % Asian, 1.1 % AIAN. DualBalance draws zero majority-minority districts. The Twin Cities slice (District 7) has the lowest non-Hispanic-white share (65.3 %) and the highest Black VAP share

(15.7 %); the remaining seven run 73–90 % non-Hispanic white. The enacted 119th Congress plan also draws zero majority-minority districts. Minnesota does not contain a region where a single minority group concentrates densely enough to support a 50 %-VAP district of conventional size.

County splits. DualBalance splits 44 of 87 counties into 143 cross-district pieces. Any line from the population centroid to the boundary crosses county lines; the enacted plan trades area balance for fewer such crossings.

What to make of this. The descriptive content of these metrics survives on a DualBalance plan; their intent-attribution role does not. §4.7 develops the distinction.

3.7 Cross-state, multi-algorithm validation

We replicated the Minnesota pipeline on five additional states spanning the density-profile and gerrymandering-history spaces: Iowa (4 districts, near-uniform density, drawn by Iowa’s nonpartisan Legislative Services Agency [7]), Massachusetts (9 districts, single-metro Boston), Texas (38 districts, polycentric), North Carolina (14 districts, the state in which *Rucho* [21] originated), and Wisconsin (8 districts, where *Whitford v. Gill* first put the Efficiency Gap [17] in front of a federal court). The data pipeline (TIGER 2020 VTDs, Census PL 94-171 demographics, dra2020/vtd.data 2020 presidential returns, TIGER 2024 cd119 spatial join for the enacted plan) is automated in `scripts/prep_state_units.py`.

To put DualBalance in context against other deterministic baselines we score two additional algorithms on the same six states. *Cascade* (`src/dualbalance/cascade.py`) is an Iowa-LSA-flavored construction: aggregate VTDs to counties, pick N farthest-spread county seeds, run capacitated first-fit county-to-district assignment, and split a county into pseudo-counties (via DualBalance internally) only when its population exceeds the per-district cap. *BDistricting* [11] is Brian Olson’s published 50-state map series, ingested via Olson’s block-level CSV output joined to our VTDs through the Census 2020 Block Assignment File (`scripts/prep_bdistricting.py`). Cascade prioritizes county integrity; BDistricting prioritizes compactness; DualBalance prioritizes area balance. Together with the enacted plan this gives four points of comparison on each state.

Table 3: Cross-state DualBalance Score. **Bold** marks the winner on each row.

State	N	DualBalance	Cascade	BDistricting	Enacted
Iowa	4	0.8132	0.8227	0.7885	0.8828
Massachusetts	9	0.7591	0.7602	0.7158	0.7246
Minnesota	8	0.6472	0.7709	0.6493	0.6391
North Carolina	14	0.7252	0.8199	0.7600	0.7689
Texas	38	0.6230	0.7943	0.6350	0.6658
Wisconsin	8	0.6556	0.8399	0.7422	0.7410

Three findings stand out.

Cascade dominates on DBS. Cascade wins on five of six states (MN, MA, TX, NC, WI) and runs second on Iowa. DualBalance wins on no state outright; BDistricting on none. Cascade’s lead comes from the structural pairing of high county integrity (1–10 splits vs. the enacted plan’s 9–30) with low area imbalance (county aggregation produces pseudo-counties of similar size across each state). The exception is Iowa, where the enacted plan is itself the result of a closely related cascade process [7] and the LSA’s manual refinement gives it a small edge.

Table 4: Cross-state Efficiency Gap (positive = R-favorable). **Bold** marks the smallest $|EG|$ on each row.

State	Statewide R %	DualBalance	Cascade	BDistricting	Enacted
Iowa	54.2	-0.173	+0.165	-0.088	+0.416
Massachusetts	32.9	-0.158	-0.158	-0.158	-0.158
Minnesota	46.4	+0.066	+0.049	+0.067	+0.068
North Carolina	50.7	+0.088	+0.026	-0.077	+0.201
Texas	52.8	+0.029	+0.009	+0.058	+0.153
Wisconsin	49.7	+0.147	+0.081	+0.142	+0.267

Table 5: Counties kept intact. Cascade splits a county only when its population exceeds the per-district cap.

State	Counties	DualBalance (split)	Cascade (split)	BDistricting (split)	Enacted (split)
Iowa	99	30	0	21	0
Massachusetts	14	12	4	10	9
Minnesota	87	44	1	25	9
North Carolina	100	56	2	37	11
Texas	254	132	10	82	30
Wisconsin	72	46	1	25	12

Different deterministic algorithms, different trade-offs. The three deterministic generators win on different axes. DualBalance minimizes area_dev on single-metro states with a clean radial structure but produces high county splits and low compactness. BDistricting maximizes compactness and pop-balance but accepts high area imbalance ($\overline{\text{area_dev}}$ near 0.8 on MA, 1.1 on TX). Cascade maximizes county integrity and area balance but tolerates looser population balance (10–20% pre-tighten). The “best” algorithm on DBS is whichever happens to weight the underlying trade-offs in a way the state’s geometry rewards.

Partisan asymmetry shrinks across every deterministic generator. On every state with nonzero statewide partisan asymmetry, all three deterministic algorithms produce smaller $|EG|$ than the enacted plan. The largest gaps appear in the two states most associated with partisan-gerrymander litigation: NC (enacted +0.20 vs. Cascade +0.03) and WI (enacted +0.27 vs. Cascade +0.08). Cascade has the smallest $|EG|$ on five of six states; BDistricting on one (Iowa). The structural claim in §4.7 is the empirical core of this finding: a generator with no political input cannot produce a +0.20 partisan-fairness metric. The result holds across all three deterministic baselines, not just DualBalance. Cascade is the existence proof of that claim under a different objective hierarchy than DualBalance’s.

4 Discussion

4.1 What the Minnesota result does and does not show

The headline (DualBalance beats the enacted Minnesota plan on the DualBalance Score) is a single-state existence result. It shows that radial slicing can score competitively on the dual-balance objective without reading party, race, or any discretionary input. It does not show that DualBalance generalizes to all 50 states, that DBS is the right scalar to optimize, or that DualBalance maps

would survive judicial scrutiny. We take those up below.

4.2 Co-design of score and algorithm

A natural objection: “ A beats B on Metric M because A and M were co-designed.” DBS (2) was specified before the algorithm choice. We tested three forms: the weighted form used here, a sum-of-means form $1/(1 + \overline{\text{pop_dev}} + \overline{\text{area_dev}})$, and an L^∞ form $1/(1 + \max_i[0.5 \text{ pop_dev}_i + 0.5 \text{ area_dev}_i])$. DualBalance beats the enacted plan on the first two; the enacted plan wins on the third, because DualBalance concentrates area imbalance in a single rural district. We use the weighted form as the headline because it cleanly implements “each district carries $1/N$ of people *and* $1/N$ of geography.” Per-district deviations are reported alongside so readers can recompute against any preferred aggregation.

The functional form satisfies four properties we wanted from the outset. (i) *Scale invariance*: the per-district inputs are relative deviations $|x - x^*|/x^*$, so the score is invariant to the units of population and area. (ii) *Equal treatment of the two objectives*: the 0.5/0.5 weighting is the unique convex combination treating $\overline{\text{pop_dev}}$ and $\overline{\text{area_dev}}$ symmetrically. (iii) *Monotonicity*: the score strictly decreases as either deviation increases. (iv) *Boundedness in $[0, 1]$* : the reciprocal form maps zero error to 1 and unbounded error toward 0, giving a calibrated reading rather than a raw cost. These do not pin DBS down uniquely, but they constrain the design space enough that the remaining choice (arithmetic mean of deviations versus a max-norm or L_p aggregation) becomes a tradeoff between average-case fairness and worst-case fairness. We report all the per-district numbers needed to compute both.

Population density variance bounds the achievable $\overline{\text{area_dev}}$ on any pop-balanced plan. Both DualBalance and the enacted plan land near $\overline{\text{area_dev}} \approx 1.0\text{--}1.1$. On Minnesota’s geometry (Twin Cities = 55 % of population in 3 % of land area) this is a hard floor. Improvements would require districts that span urban and rural, which is exactly what radial slicing does. The 8.7-point gap (103.9 vs. 112.6) is the operational measure of how much area balance DualBalance recovers given that floor.

4.3 Compactness is the trade

DualBalance trades compactness for area balance by design. This is not an incidental cost we apologize for: it is the mechanism. A district that holds $1/N$ of a high-variance density distribution must either span the full density range (and look elongated) or sit inside a single density regime (and produce an unbalanced area share). Hand-drawn maps typically choose the second; DualBalance chooses the first. The PP=0.047 worst-slice number is what choosing the first looks like on Minnesota’s geometry. The enacted Minnesota plan’s worst PP is 0.178, roughly four times higher, and its $\overline{\text{area_dev}}$ is correspondingly 8.7 points worse.

Three observations qualify the trade.

Compactness is a metric, not a constitutional requirement. *Polsby-Popper* as a measure was published in 1991 [14]; it appears in no federal statute and binds no court. Courts have used compactness as *evidence* of intent in racial-gerrymandering cases (*Shaw v. Reno* and progeny), but the constitutional violation in those cases turns on the use of race in line-drawing, not on the resulting shape per se. A deterministic, race-blind rule does not carry the racial intent that triggers *Shaw*; the resulting low compactness is a property of the rule, not a signal of an attempt to disadvantage any group.

Compactness assumes a baseline of comparison. A district is “bizarrely shaped” relative to surrounding districts that are not. If all eight districts in a state are produced by the same geometric rule and all eight share a similar slice-like character, the comparative baseline shifts. The aesthetic objection to long thin shapes is partly historical: it picks out districts that depart from the surrounding norm of compact blobs. A scheme in which every district is a slice does not pick anything out.

The Voting Rights Act trade-off is real. *Allen v. Milligan* [22] reaffirmed Section 2’s effects-based vote-dilution doctrine; *Alexander* [24] and *Callais* [25] narrowed its practical reach. Section 2 asks about effects on minority voters’ opportunity to elect, not about intent. DualBalance is race-blind, but blindness to race does not satisfy Section 2 by itself. In jurisdictions where the *Gingles* [19] preconditions are met, DualBalance may produce a plan that fails Section 2 review even though no one drew lines to dilute. The legislative question (whether to accept reduced minority opportunity in exchange for a generator that cannot be partisan-tuned) is for legislatures and courts, not for the algorithm.

4.4 Relationship to prior deterministic methods

DualBalance differs from each prior method on a specific axis.

Versus Smith’s Shortest Splitline [16]. Splitline recursively bisects the state with the shortest population-balancing line, with no global compactness or balance criterion. The cuts are visually striking but arbitrary: a district can lose half its territory because the shortest cut happens to land there. DualBalance grounds its geometry in the population centroid, so the slices reflect the state’s density structure.

Versus Olson’s BDistricting [11]. BDistricting minimizes population-weighted distance to dispersed seed centers under equal-population, with Lloyd-style iteration to convergence. The resulting maps are blob-shaped and score well on compactness but exhibit large area imbalance. DualBalance differs in two ways: seeds are radial-clustered around the centroid (not dispersed), and there is no iteration. On the six-state comparison (§3.7) BDistricting wins on no state outright on DBS, but produces the smallest |EG| on Iowa and the most-compact districts in every state.

Versus centroidal power diagrams [3]. The Cohen-Addad-Klein-Young construction solves for additive weights $\{w_i\}$ so each power-diagram cell carries exactly $1/N$ of the mass. It balances one quantity (population) exactly. DualBalance balances two (population and area) approximately, with the second recovered from the radial seed geometry rather than from the optimization.

Versus capacitated k -means [8] and Hess-style location-allocation [6, 10]. The capacitated assignment step (§2.3) is Hess’s 1965 transportation step. DualBalance’s novelty is the seed placement: radial seeds produce a slice geometry that Lloyd iteration would destroy, because re-centering pulls seeds away from the tight radial cluster. Disabling iteration is what preserves the geometry.

Versus our own Cascade baseline. Cascade (§3.7, `src/dualbalance/cascade.py`) is an Iowa-LSA-flavored construction we include in the empirical comparison: it aggregates VTDs to counties, picks N farthest-spread county seeds, runs capacitated first-fit assignment at county granularity,

and splits a county into pseudo-counties (via DualBalance internally) only when its population exceeds the per-district cap. Cascade beats DualBalance on DBS on all six states tested, sometimes by wide margins (MN +0.12, TX +0.17, NC +0.09, WI +0.18); on the six-state set Cascade is the stronger *deterministic baseline* against the enacted plan. We include both algorithms because they make the conceptual contribution sharper: DualBalance and Cascade prioritize different objectives (radial slicing for density mixing; county integrity for administrative coherence) and produce different maps on every state, yet both are race-blind, partisan-blind, and deterministic by construction. The forensic critique in §4.7 applies equally to either, and to BDistricting, and to any future deterministic generator built on similar principles.

4.5 Limitations

Six-state evidence, not fifty-state generality. Three deterministic algorithms (DualBalance, Cascade, BDistricting) have been run end-to-end on six states (§3.7). On DBS, Cascade wins on MN, MA, TX, NC, and WI; the enacted plan wins only on Iowa. Across all six states with nonzero partisan asymmetry, every deterministic algorithm produces a smaller $|EG|$ than the enacted plan, with the largest gaps in the gerrymander-prone states. The remaining 44 states have not been run, and six states does not establish generality across the density-profile or litigation-history spaces.

Worst-district area outlier. DualBalance concentrates the area imbalance in a single rural district (here, District 2 at 275 % over-target). This is geometric, not algorithmic: any pop-balanced partition of a state with Minnesota’s density profile must give some rural district disproportionate area. DualBalance’s $\text{area_dev}_{\max}$ of 275 % is worse than the enacted plan’s 241 % even though the mean is lower, so DualBalance distributes the imbalance unevenly. Whether evenly distributed area imbalance is preferable to a single concentrated outlier is a separable normative question.

No multi-unit transportation step. The optional `--tighten-pop` pass is a greedy single-unit swap. A true 2D transportation LP bounding both population and area would directly minimize the DBS objective; we leave that to future work.

Computational cost. Pure DualBalance runs in under a second on Minnesota. The tightening pass takes ~ 18 s for 80 swaps. We have not profiled California or Texas; worst-case scaling is $O(|U|^2 N)$ per swap due to the contiguity check.

The compactness floor. The worst slice’s $PP = 0.047$ is well below the informal court-testimony threshold of 0.10. We argue in §4.3 that the *Shaw* [20] doctrine turns on intent rather than shape per se, but readers who reject that argument should regard the 0.047 figure as a hard ceiling on legal viability.

4.6 Future work

Full-50-state validation. Six states (MN, IA, MA, TX, NC, WI) have been run (§3.7). Extending to the remaining 44 requires no algorithmic change. California, Florida, Pennsylvania, and Ohio are the highest-value next targets: California for its size and many metros, Florida for its peninsula geometry, Pennsylvania and Ohio for their roles in state-court partisan-gerrymandering jurisprudence.

Multi-center radial seeding. For polycentric states, place seeds on circles around *each* of k population centers, with k chosen deterministically from the density distribution. Not yet implemented.

Direct 2D transportation step. Replace the DualBalance-plus-tighten two-stage pipeline with a single optimization that minimizes DBS under contiguity constraints. More expensive than the current pipeline; the mathematically clean answer to “which plan directly minimizes the DualBalance objective.”

Splitline benchmark. The six-state benchmark (§3.7) covers DualBalance, Cascade, BDistricting, and the enacted plan. Adding Smith’s Shortest Splitline [16] would give a fourth deterministic baseline; computing it from scratch on each state is straightforward but has not yet been done.

Scoring-harness extensions. The current harness reports population, area, Polsby-Popper, and Reock. Natural additions include partisan diagnostics (efficiency gap, mean-median, declination) on voter-registration data, county-split counts, and minority representation diagnostics under Section 2. These are reporting extensions and would not feed back into the generator (§2, “out-of-scope inputs”).

4.7 Forensic metrics applied to a generative procedure

Deterministic generation has interpretive consequences for the fairness-metrics literature. Most gerrymandering diagnostics carry two functions. They *describe* a property of the plan (a shape, a vote distribution, a count of split counties). And they support an *inference* about the line-drawer’s motive (that the shape was chosen, that the votes were packed and cracked, that the splits followed a partisan or racial pattern). The two are nearly always entangled in practice because the literature was built around enacted plans, where intent is always at issue.

DualBalance separates them. The descriptive function survives intact: a PP of 0.09 still names a non-compact shape, with administrative and legitimacy consequences; an Efficiency Gap of +6.6 % still names a real wasted-vote asymmetry, with representational consequences; zero majority-minority districts still names a partition under which no minority group reaches a 50 % VAP share, with Section 2 consequences. The inferential function does not. DualBalance is a pure function of geometry, population, and seat count, so the chain from observed property back to motive has no actor to terminate on.

Three implications, narrowed accordingly.

Shaw-style shape inference loses purchase, but compactness still matters administratively. *Shaw v. Reno* [20] uses bizarre shape as evidence that race predominated in line-drawing. The inference chain runs shape $\rightarrow$ unusual process $\rightarrow$ illegitimate criterion $\rightarrow$ race. On DualBalance the first link is broken: every district produced by the rule has the same shape character, so unusual shape no longer implies unusual process. *Shaw* does not reach a race-blind algorithm. But compactness still has a non-inferential role. Districts that are hard to identify with on a map, hard to administer, or hard to campaign in carry costs independent of who drew them. DualBalance’s PP=0.047 worst slice is a real cost. The argument here is not that compactness ceases to matter, only that it ceases to be evidence of motive.

Partisan metrics describe effects, not intent. *Rucho v. Common Cause* [21] foreclosed federal partisan-gerrymandering review for lack of a manageable standard for distinguishing acceptable from unacceptable partisan motivation. The question is moot on DualBalance (there is no motivation), but the partisan *effect* the metrics measure is not. An Efficiency Gap of +6.6% on a DualBalance map still corresponds to wasted-vote asymmetry that affects representation; state-court tests that interpret EG as evidence of effect, rather than as evidence of intent, still bind. The rhetorical move from “the map shows EG +6.6%” to “someone built that EG +6.6%” becomes unavailable on DualBalance, but the metric continues to measure what it measured.

Ensemble outlier tests lose their baseline. The Duke and MGGG ensembles [4, 5] compare an enacted plan to a sample of legally compliant alternatives a neutral line-drawer might plausibly have produced. The sample stands in for the absent neutral line-drawer; the enacted plan being an outlier suggests the actual line-drawer differed from the neutral one. DualBalance *is* the neutral line-drawer in a stronger sense than the ensemble’s sample mean. Asking whether the DualBalance plan is an outlier against its own ensemble may not be meaningful: the ensemble collapses to one point. Ensembles can still play a comparison role across *different* algorithmic generators (e.g. DualBalance vs. BDistricting vs. splitline), but the question is no longer “is this map suspiciously different from neutral?”

Metrics remain informative as descriptive summaries of plan effects. A DualBalance plan with $PP_{\min} = 0.047$, Efficiency Gap +6.6%, and zero majority-minority districts has low compactness, an R-favoring wasted-vote asymmetry, and no MMD. Those facts carry administrative, representational, and Section 2 consequences. The inference from those facts to a person who chose them is what deterministic generation removes.

4.8 What this paper claims, and what it does not

The paper makes four claims.

First, that a knob-free, race-blind, partisan-blind, deterministic districting algorithm exists that produces output competitive with, and on a single empirically tested state superior to, the corresponding hand-drawn enacted plan on a prespecified population-and-area objective. The algorithm’s output is byte-reproducible and updates only with the ten-year census.

Second, that the algorithm’s procedural neutrality is symmetric. The rule cannot be used to harm any group, party, or incumbent, and cannot be used to help any of them. The same property that forecloses a partisan gerrymander forecloses a race-conscious remedy. The symmetry is the algorithm’s defining property; whether it is morally preferable to the status quo of partisan and remedial discretion is a question for legislatures, courts, and voters.

Third, that the post-*Callais* legal landscape gives a generator of this kind a narrower constitutional exposure than any race-conscious or partisan-conscious alternative. Federal partisan review is foreclosed under *Rucho*; federal racial-gerrymander risk is contracting under *Alexander* and *Callais*; state-court remedies for partisan claims depend on the politics of each state’s supreme court. A rule whose inputs do not contain race or party is, in the strict sense, not subject to either body of doctrine.

Fourth, the conventional gerrymandering metrics (compactness scores, Efficiency Gap, median, ensemble outliers, majority-minority counts) play two roles on enacted plans, only one of which transfers to DualBalance. As descriptions of plan effects (shape, wasted-vote asymmetry, minority opportunity) they remain valid and are reported alongside the DualBalance Score. As evidence of line-drawer intent they do not transfer. The intent-attribution use of these metrics,

not their measurement use, is what the generative case breaks. This is the most contested framing claim in the paper.

The paper does not claim that DBS is universally the right objective, that DualBalance dominates enacted plans on all states, that compactness can be ignored, that effects-based fairness analyses (including Section 2) become irrelevant, or that the metric toolkit developed over the past thirty years is wrong. The narrow claim is that the intent reading of those metrics presupposes a line-drawer who is not present in this construction.

Whether deterministic generation is preferable to discretionary redistricting is a political and constitutional question; the algorithm itself cannot answer it.

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